

EXPERIMENT - 19

ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

CODE:

```
// Program Sketch

int trig = 10;

int echo = 9;

long duration;

int cm;

void setup() {
// put your setup code here, to run once:
pinMode(trig, OUTPUT);
pinMode(echo, INPUT);
Serial.begin(9600);
}

void loop() { // Sketch to calculate ultrasonic distance
digitalWrite(trig, LOW);
delayMicroseconds(10);
digitalWrite(trig, HIGH);
delayMicroseconds(10);
digitalWrite(trig, LOW);
delayMicroseconds(10);
duration = pulseIn(echo, HIGH);
cm = (duration/2) * 0.034;
// Print Value on Serial Monitor
Serial.print(" Distance = ");
```

```

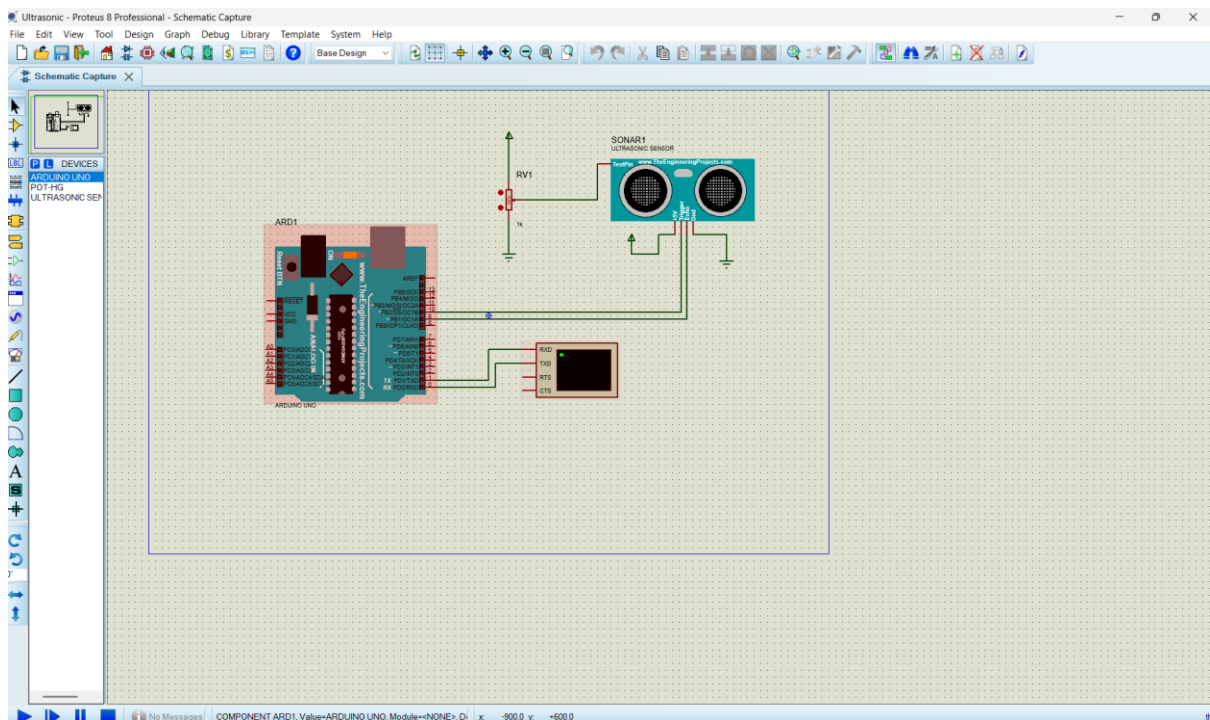
Serial.print(cm);

Serial.println(" cm");

}

```

SIMULATION:



The screenshot displays the Proteus 8 Professional Schematic Capture interface. The main workspace shows a circuit diagram with an Arduino Uno (ARD1) and a SONAR1 Ultrasonic Sensor. The sensor is connected to a 5V supply through a 1k resistor (RV1). The Arduino's TXD, RTS, and CTS pins are connected to a Virtual Terminal window. The terminal window displays the output: "Distance ~ 550 cm". The left sidebar shows the "TERMINALS" list, and the bottom status bar indicates "ANIMATING 00:00:05.050000 (CPU load 40%)".